

EXPERIMENT - 1A

Smallest of 2 Numbers

AIM:

To write an assembly language Program to find the smallest number in an array using 8085.

SOFTWARE USED:

GNU sim 8085.

ALGORITHM:

- 1) Initialize the count.
- 2) Get the input numbers.
- 3) Compare the content of A with HL pair for all input numbers.
- 4) Store the smallest number in output.
- 5) End.

Program:

Memorics:

```
LXI H, 8000H ; Load HL register pair with 8000H
MOV B, M ; Move content of HL into B
INX H ; Increment HL pair
MOV A, M ; Move HL 8001H into A
CMP B ; Compare B with A
JC store ; Jump to label
store: ; Flag is set (ALB)
```

MOV A 13 ; if carry flag is not set (A >= B), load into register B
 STORE: STA 8010H ; store content of A into 8010H

Halt the program execution.

INPUT

Address	Data
8000	10

Output

Address	Data
8010	3

RESULT:

Thus the assembly language program to find the smallest number in an array is performed.



Registers

A 00
BC 00 00
DE 00 00
HL 00 00
PSW 00 00
PC 42 10
SP FF FF
Int-Reg 00

Flag

S 0
Z 1
AC 0
P 1
C 0

Load me at

```
1 LDA 8000H
2 MOV B, A
3 LDA 8001H
4 CMP B
5 JC STORE
6 MOV A, B
7 STORE: STA 8010H
8 HLT
```

Decimal - Hex Conversion

Decimal

Hex

0

0

→ To Hex

← To Dec

I/O Ports

0

-

+

00

Update Port Value

Memory

8001

-

+

10

Update Memory

Data Stack KeyPad Memory I/O Ports

Start 8010

OK

Address (Hex) Address Data

1F4A	8010	5
1F4B	8011	0
1F4C	8012	0
1F4D	8013	0
1F4E	8014	0
1F4F	8015	0
1F50	8016	0
1F51	8017	0
1F52	8018	0
1F53	8019	0
1F54	8020	0
1F55	8021	0
1F56	8022	0
1F57	8023	0
1F58	8024	0
1F59	8025	0

Line No Assembler Message

0 Program assembled successfully

Simulator: Idle

